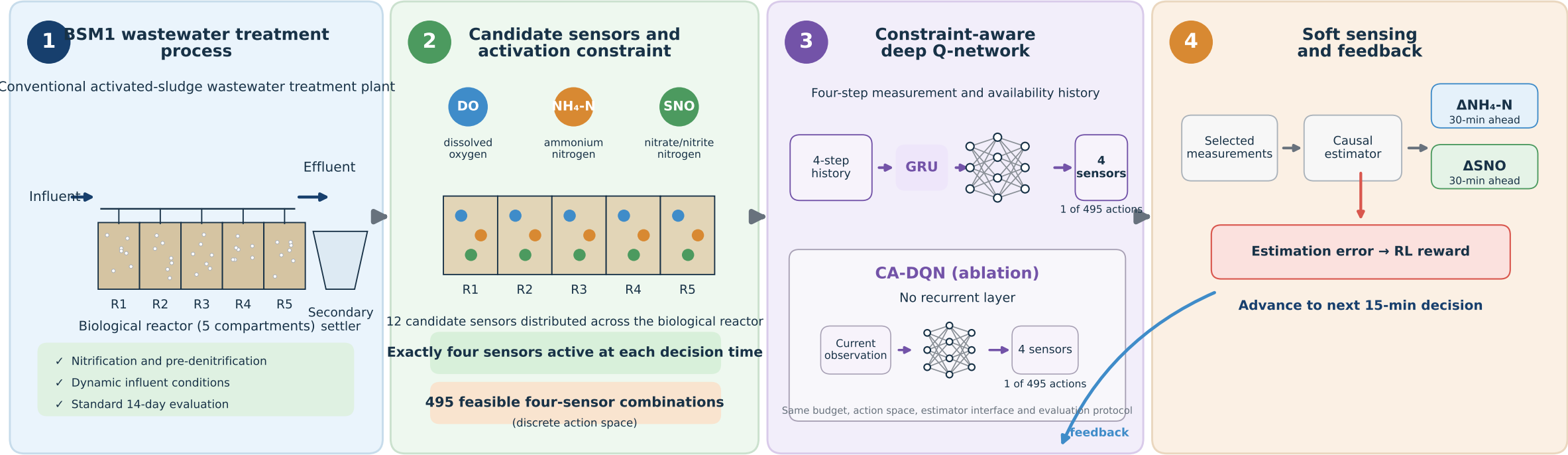


Constraint-Aware Deep Q-Network for Dynamic Sensor Activation Scheduling in Wastewater Treatment Plants: A Multi-Seed Robustness and Ablation Study

SMARTER MONITORING | EFFICIENT SENSOR USE | ROBUST EVALUATION



Key findings



Robust performance

Consistent results across 10 independent training seeds



Effective sensor selection

Exactly 4 of 12 sensors with 495 feasible configurations



Temporal variation

Informative sensor configurations vary across evaluation windows



Recurrence ablation

CA-RDQN and CA-DQN compared under the same protocol

Constraint-aware reinforcement learning for efficient, reliable and sustainable wastewater monitoring